

CoRoT-1b

R-2235

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_250104 · created 2026-08-15T01:38:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460679.6964 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.041
Transit depth (Rp/Rs)^2	2.04 +/- 1.16 [%]
Orbital Inclination (inc)	84.83 +/- 2.89
Airmass coefficient 1 (a1)	0.78 +/- 0.13
Airmass coefficient 2 (a2)	0.146 +/- 0.068
Scatter in the residuals of the lightcurve fit is	16.97 %
Best Comparison Star	#1 - [512, 226]
Optimal Aperture	1.3
Optimal Annulus	12.59
Transit Duration (day)	0.103 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.041 vs 0.1388 (deviation 0.09σ)
Duration fit vs published	0.103 d vs 0.1046 d (deviation 0.12σ)
Mid-time O–C	-14.2 min
Depth SNR	3.1
Residual scatter	16.97 %

Light curve

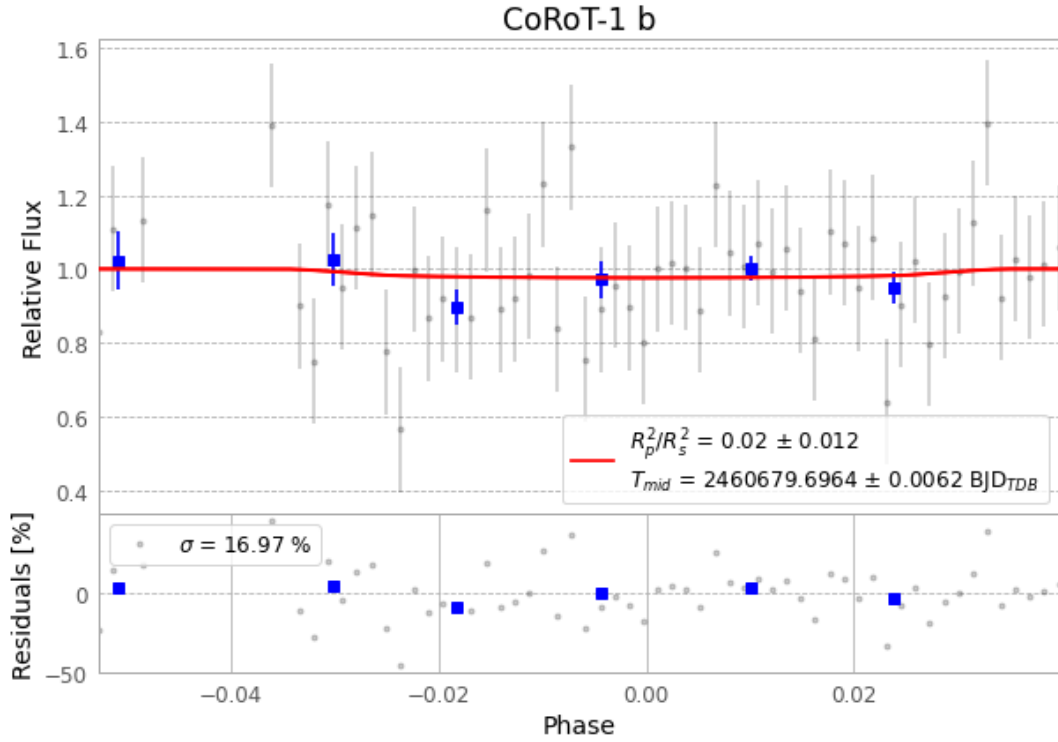


Figure 1 — FinalLightCurve_CoRoT-1 b_2025-01-03.png (SHA-256 138070eadb14130b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 168], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 745bfe7cc36b463e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

68 FITS frames (45 MB), CoRoT-1250104023916.FITS → CoRoT-1250104060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-250104.log (649af84ec013...), FinalLightCurve_CoRoT-1 b_2025-01-03.png (138070eadb14...), FinalLightCurve_CoRoT-1 b_2025-01-03.csv (4f9aaf1f4f7b...)

Compute resources

Wall time 158.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:38Z.